

Oral Cavity

Tongue & Palate

second part

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Oral Cavity (Mouth)

1) Tongue.

2) Palate.

2) Palate.

1) Hard Palate.

2) Soft Palate.

3) Palatine Aponeurosis.

4) Muscles of soft palate.

5) Sensory Nerve Supply

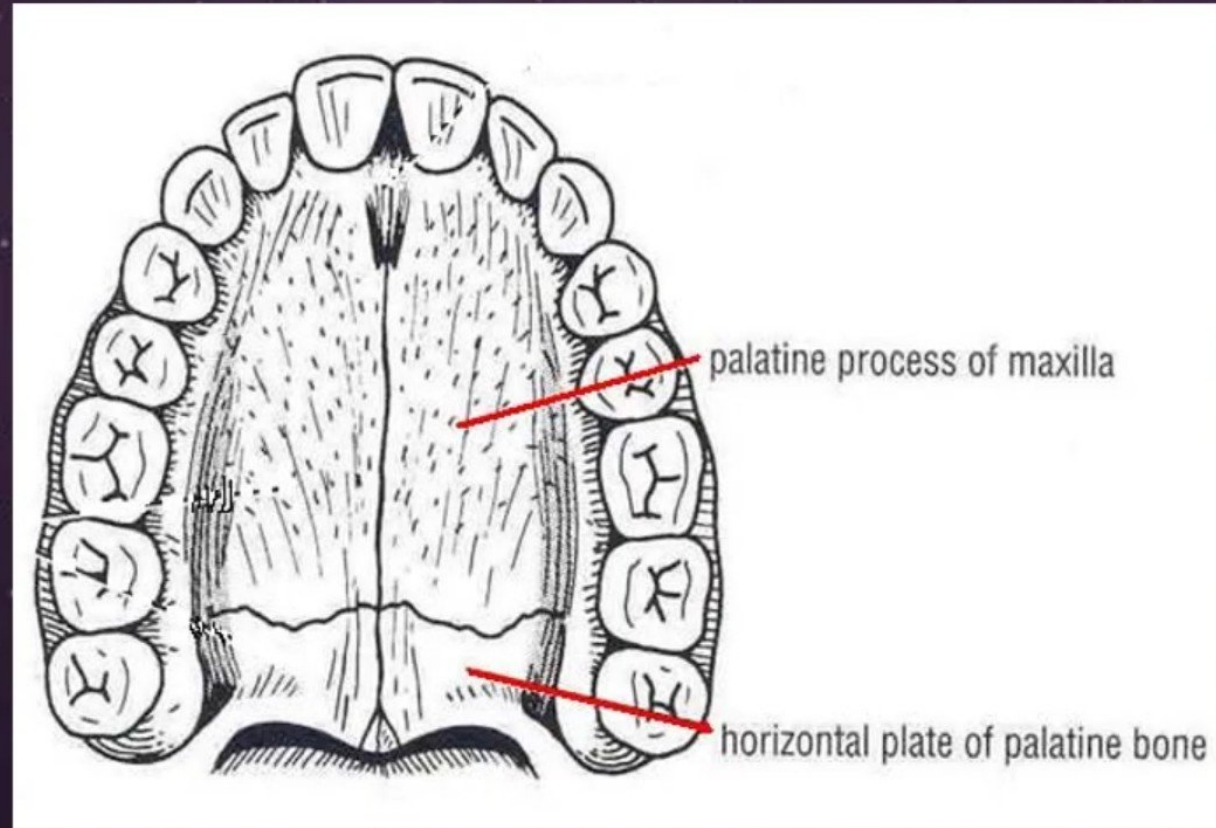
6) Motor Nerve Supply

7) Blood Supply

8) Clinical Notes

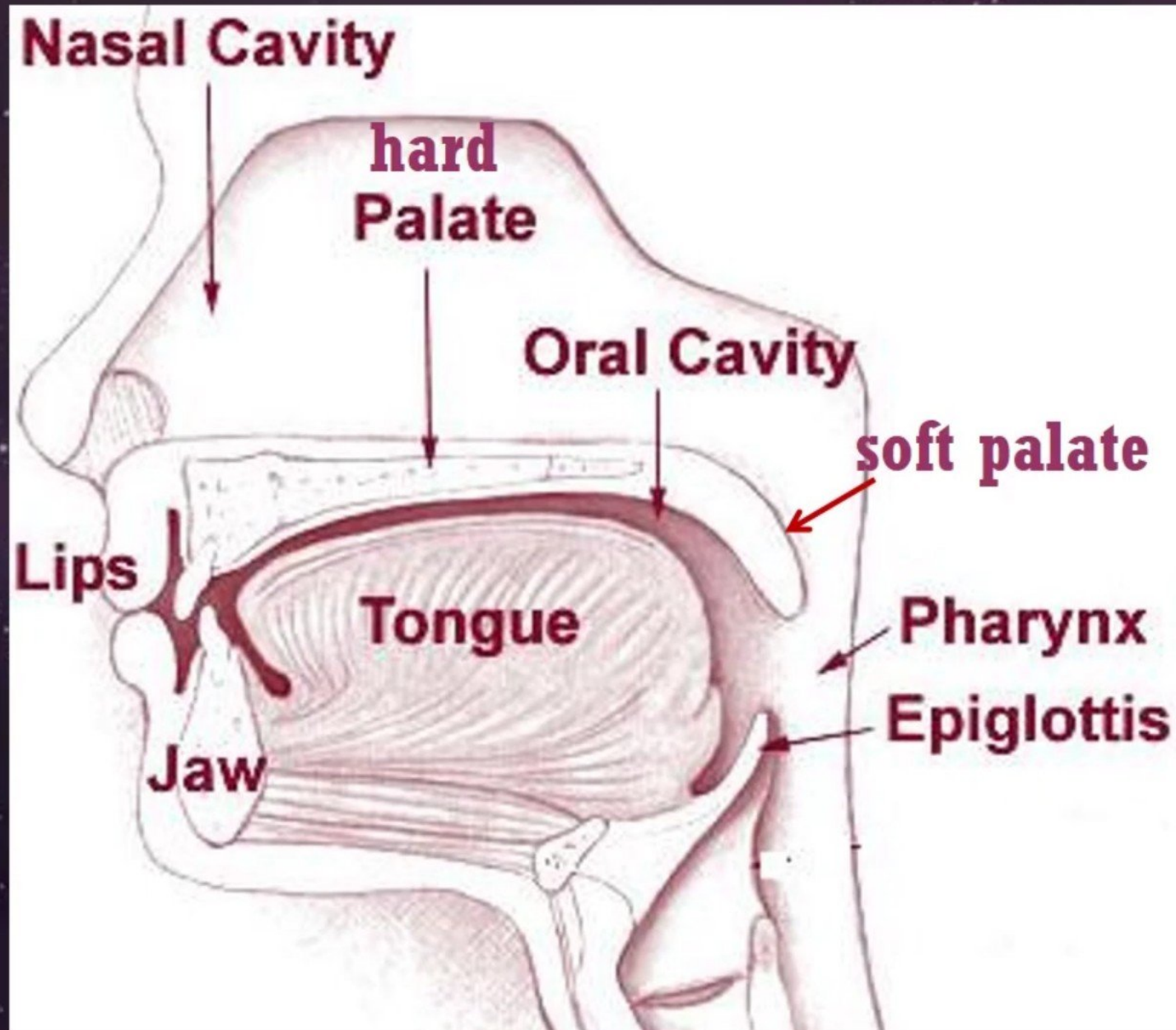
Hard Palate

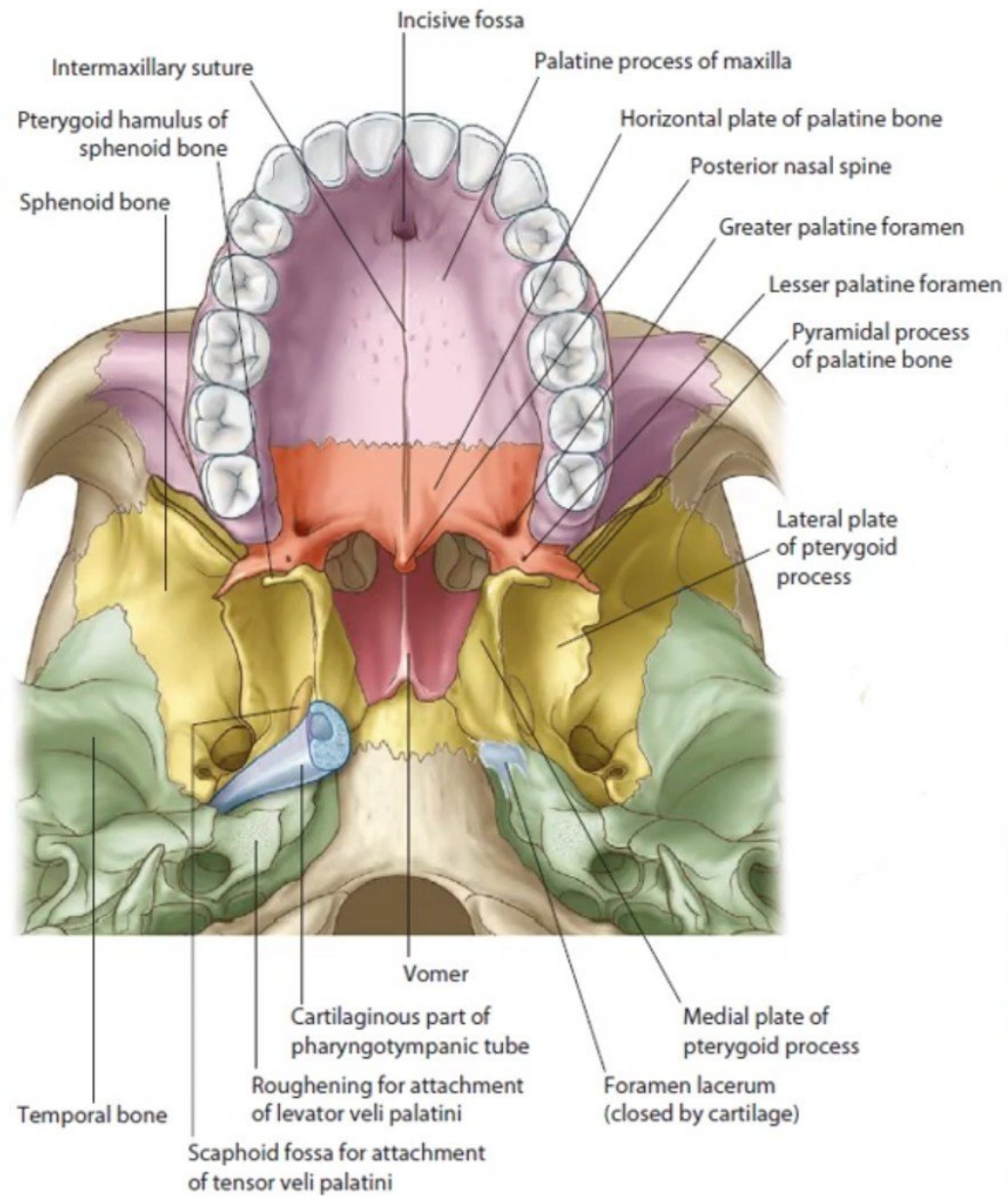
- ◉ Lies in the **roof of the oral cavity**
- ◉ Forms the **floor of the nasal cavity**
- ◉ Formed by:
 - ✎ Palatine processes of maxillae in front
 - ✎ Horizontal plates of palatine bones behind
- ✎ Bounded by **alveolar arches**



Palate

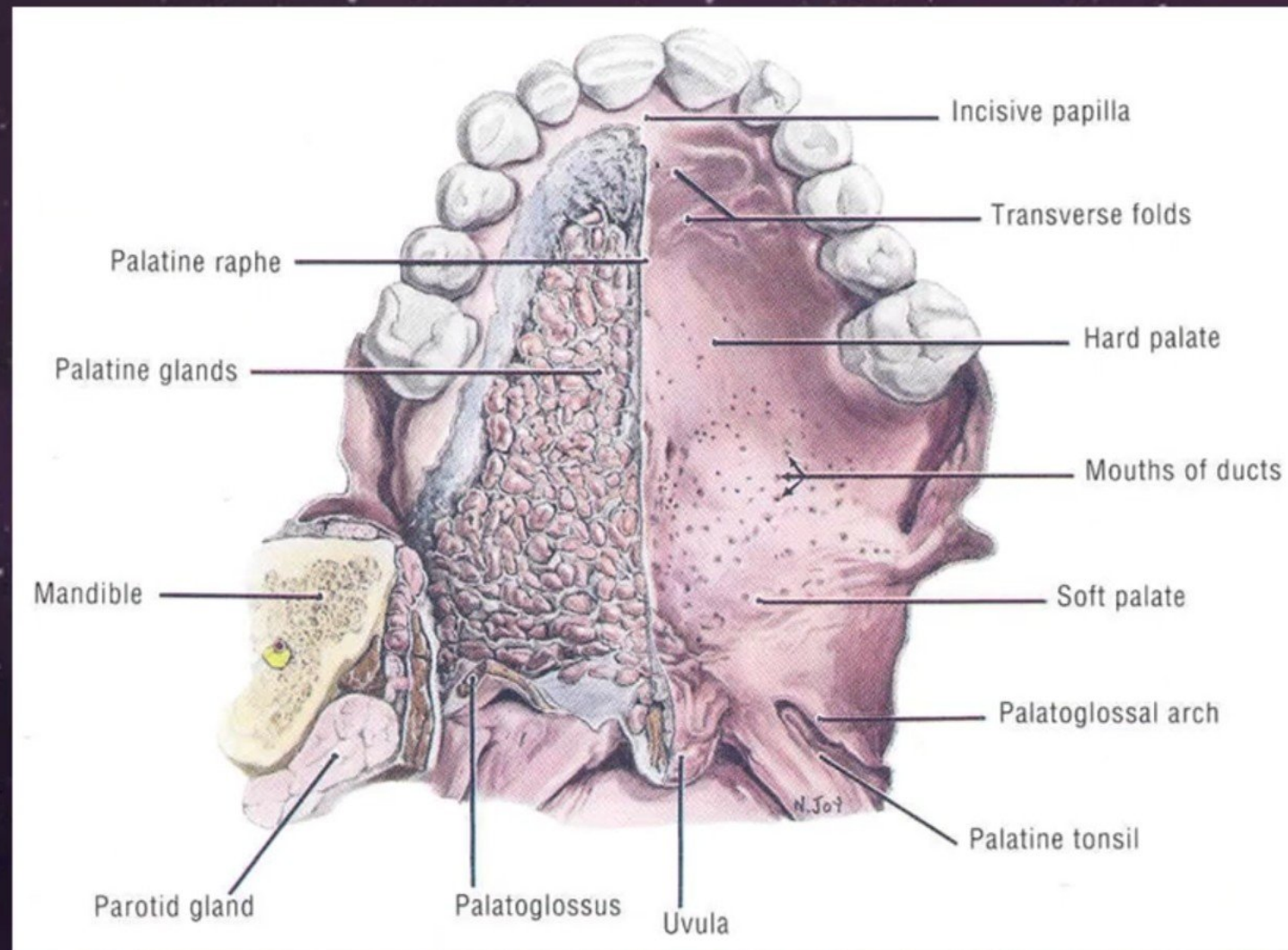
- Lies in the **roof** of the oral cavity
- Has two parts:
 - ✧ **Hard (bony)** palate anteriorly
 - ✧ **Soft (muscular)** palate posteriorly





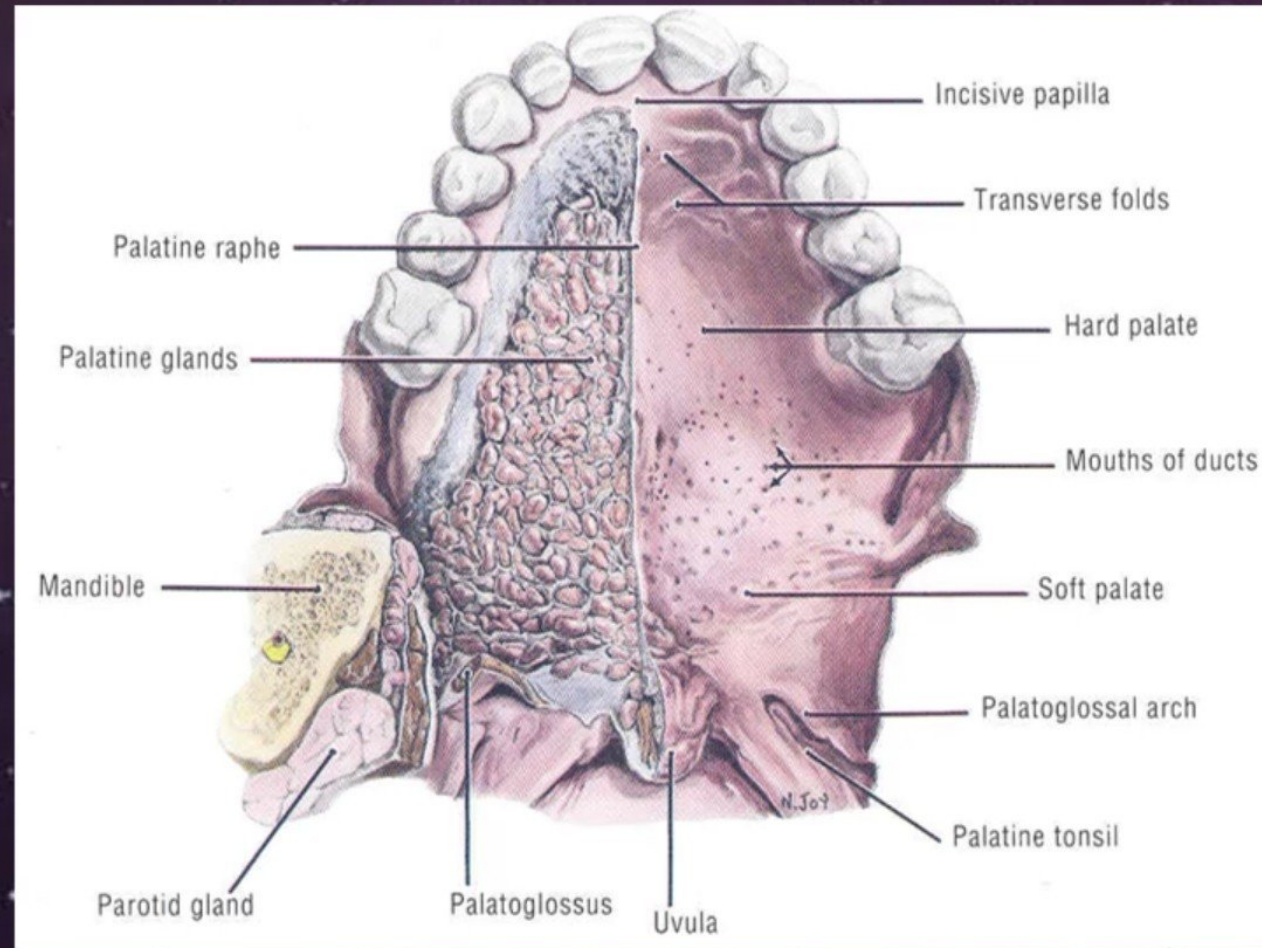
Hard Palate

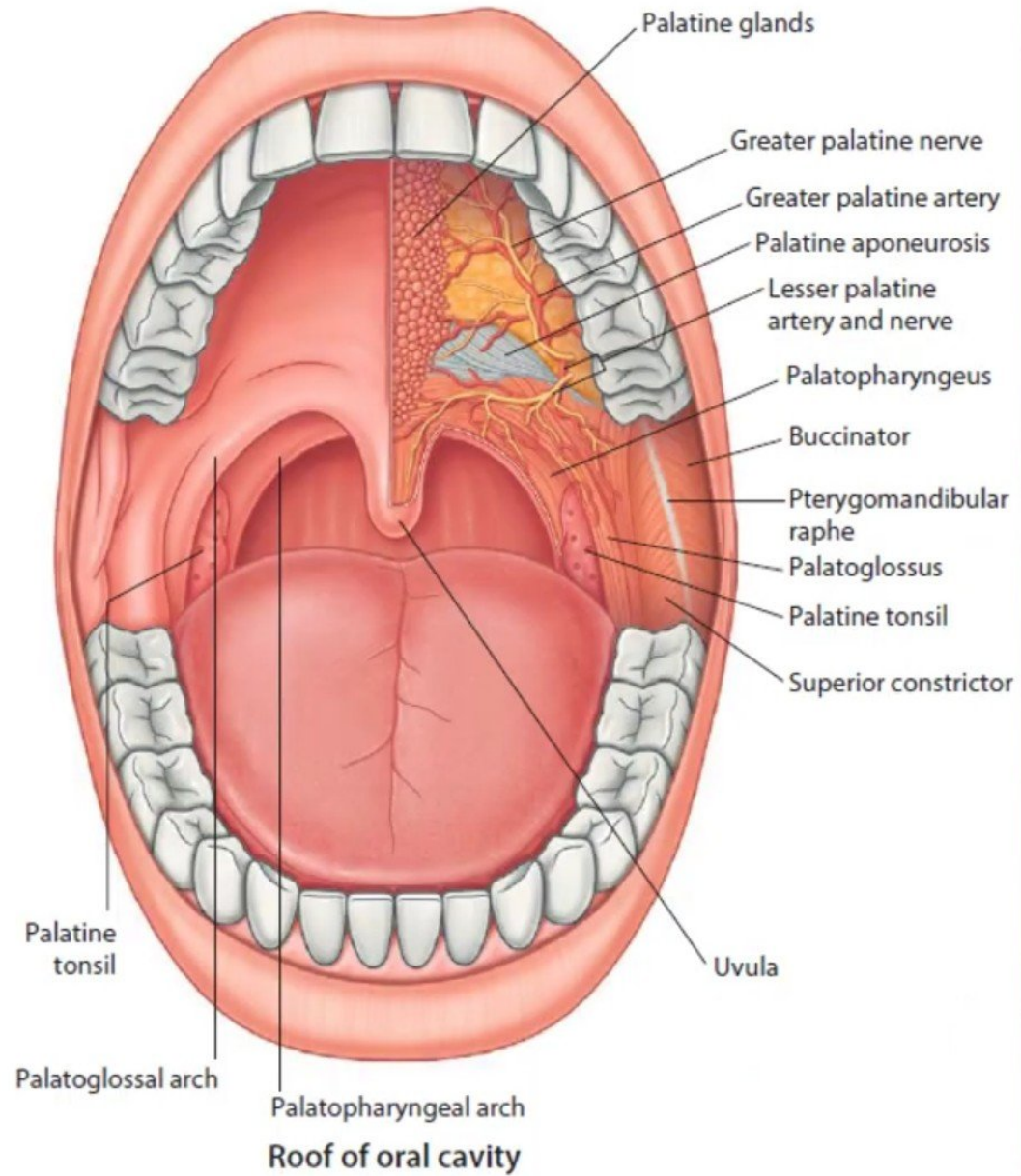
- Posteriorly, continuous with soft palate
- Its undersurface covered by **mucoperiosteum**
- Shows transverse ridges in the anterior parts



- ◉ Attached to the posterior border of the hard palate
- ◉ Covered on its upper and lower surfaces by **mucous membrane**
- ◉ Composed of:
 - ✎ Muscle fibers
 - ✎ An aponeurosis
 - ✎ Lymphoid tissue
 - ✎ Glands
 - ✎ Blood vessels
 - ✎ Nerves

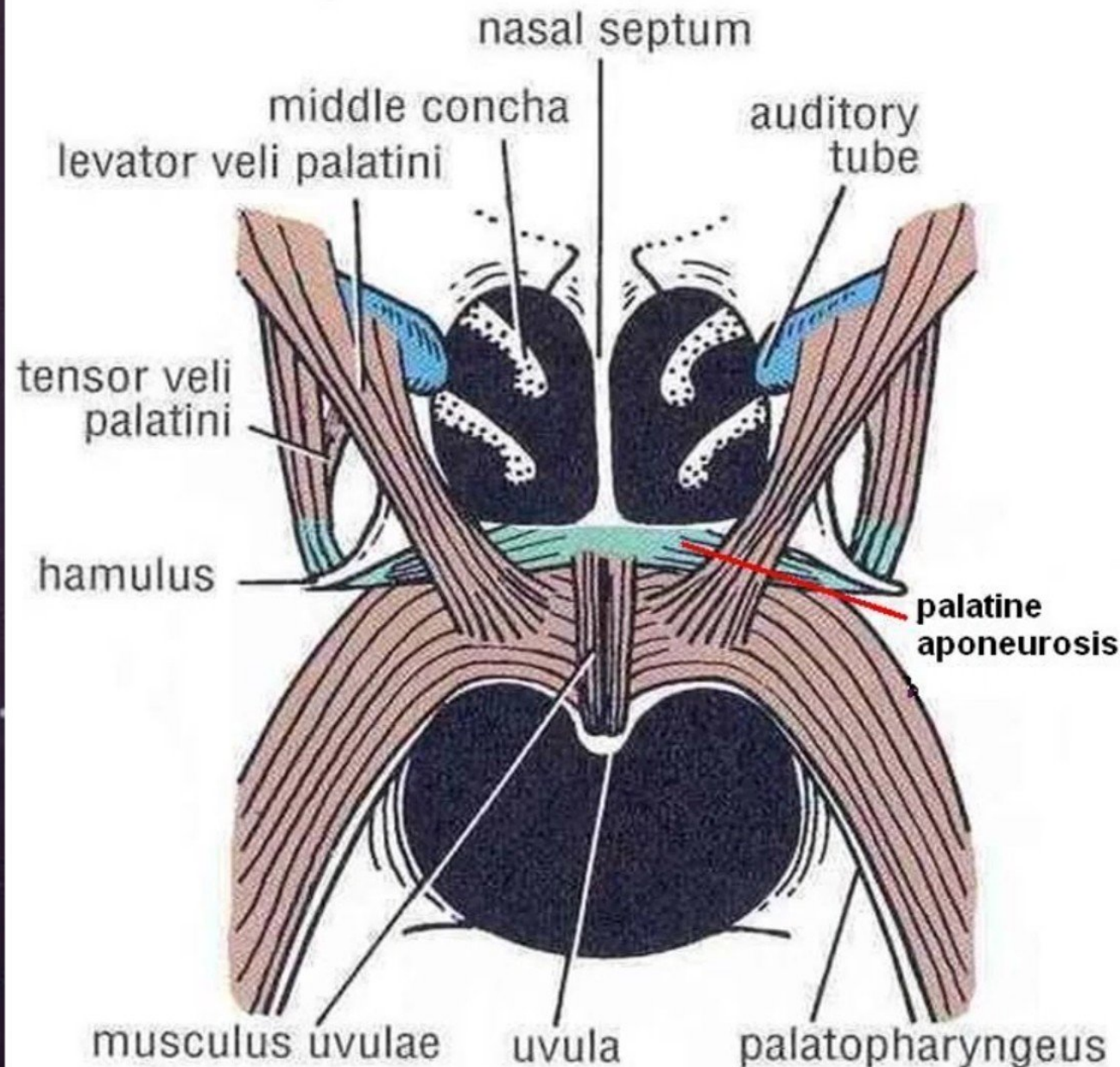
Soft Palate



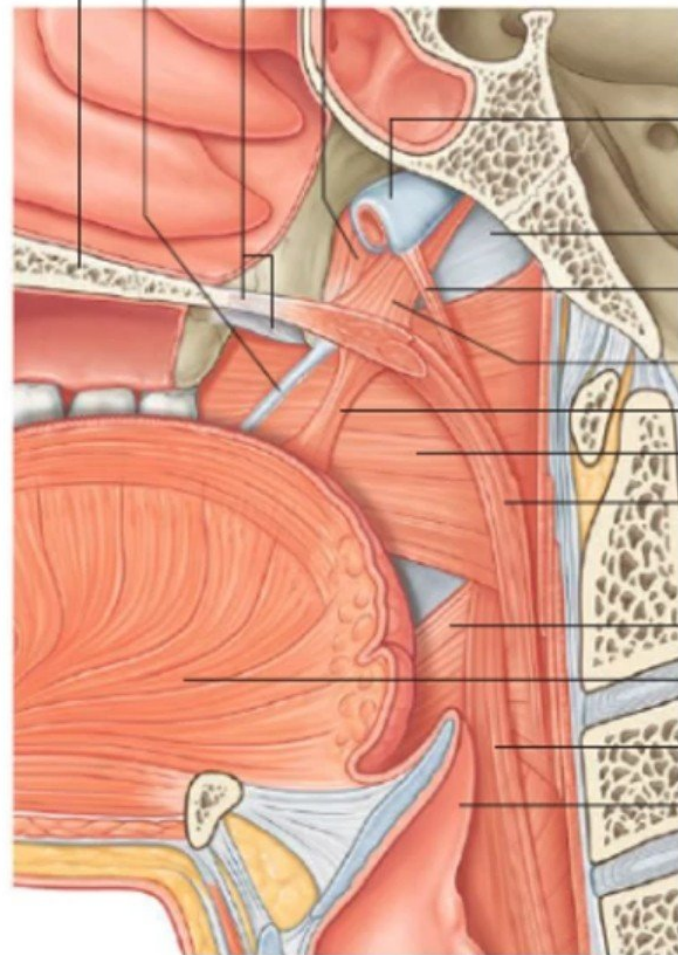


Palatine Aponeurosis

- ◉ Fibrous sheath.
- ◉ Attached to posterior border of hard palate.
- ◉ Is expanded tendon of **tensor velli palatine**.
- ◉ Splits to enclose **musculus uvulae**.
- ◉ Gives origin & insertion to palatine muscles.



Pterygomandibular raphe
Hard palate
Palatine aponeurosis and tendon of tensor veli palatini
Tensor veli palatini



Muscles of the soft palate (sagittal section)

Cartilaginous part of pharyngotympanic tube

Pharyngobasilar fascia

Salpingopharyngeus

Levator veli palatini

Palatoglossus

Superior constrictor

Palatopharyngeus

Musculus uvulae

Middle constrictor

Genioglossus

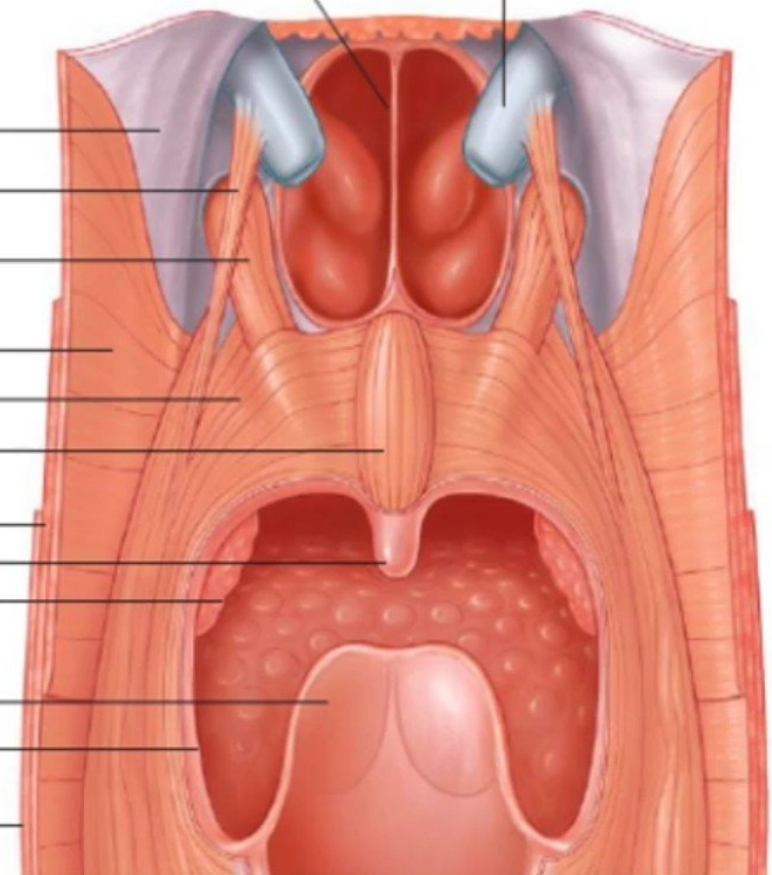
Stylopharyngeus

Epiglottis

Pharyngo-epiglottic fold

Inferior constrictor

Nasal septum
Cartilaginous part of pharyngotympanic tube



Muscles of the soft palate (posterior view)

Muscles

⊙ Tensor veli palatini

✂ **Origin:** spine of sphenoid; auditory tube

✂ **Insertion:** forms palatine aponeurosis

✂ **Action:** Tenses soft palate

✂ Levator veli palatini

✂ **Origin:** petrous temporal bone, auditory tube, palatine aponeurosis

✂ **Insertion:** palatine aponeurosis

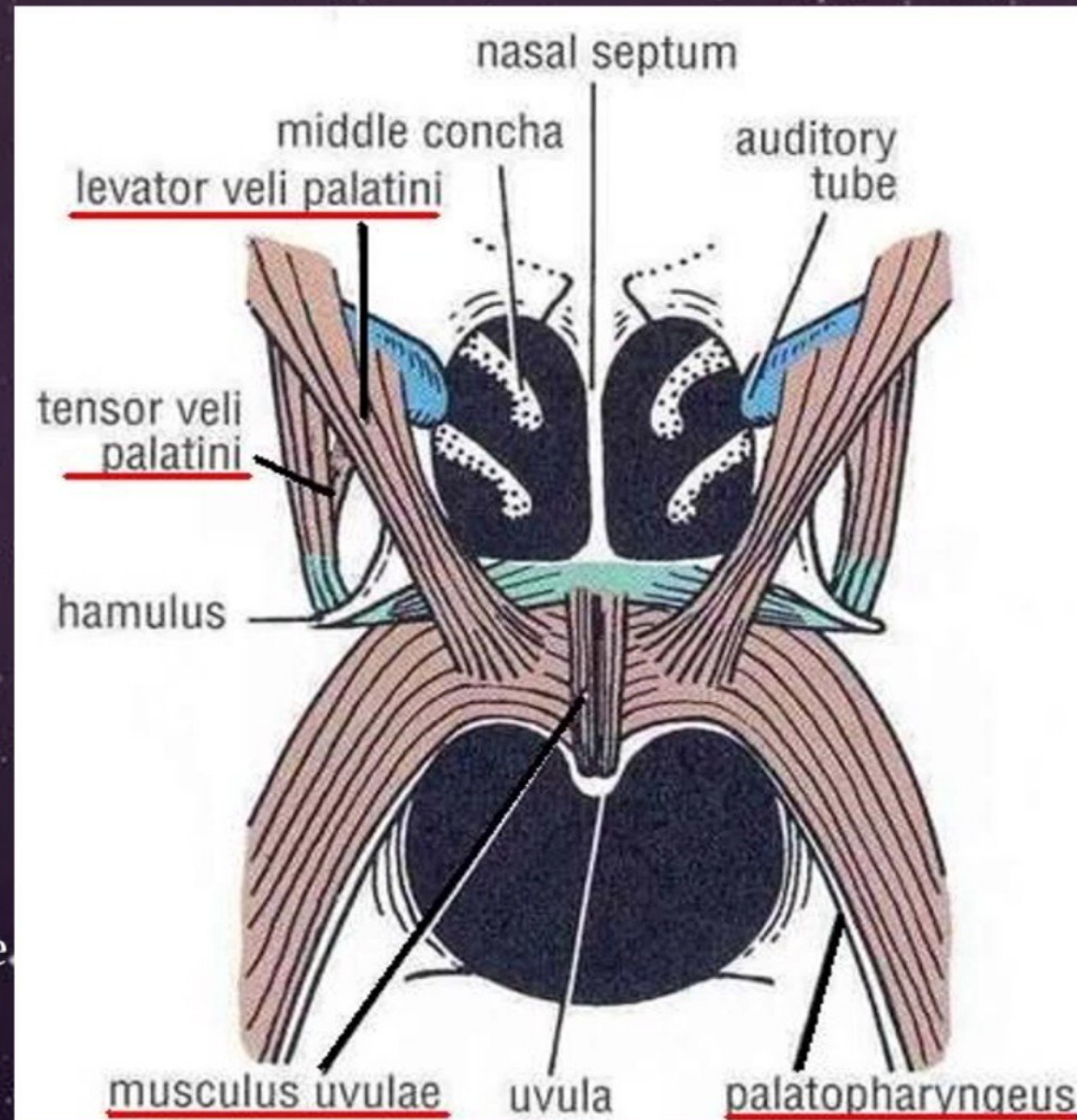
✂ **Action:** Raises soft palate

✂ Musculus uvulae

✂ **Origin:** posterior border of hard palate

✂ **Insertion:** mucosa of uvula

✂ **Action:** Elevates uvula



Muscles

& Palatoglossus

✧ **Origin:** palatine aponeurosis

✧ **Insertion:** side of tongue

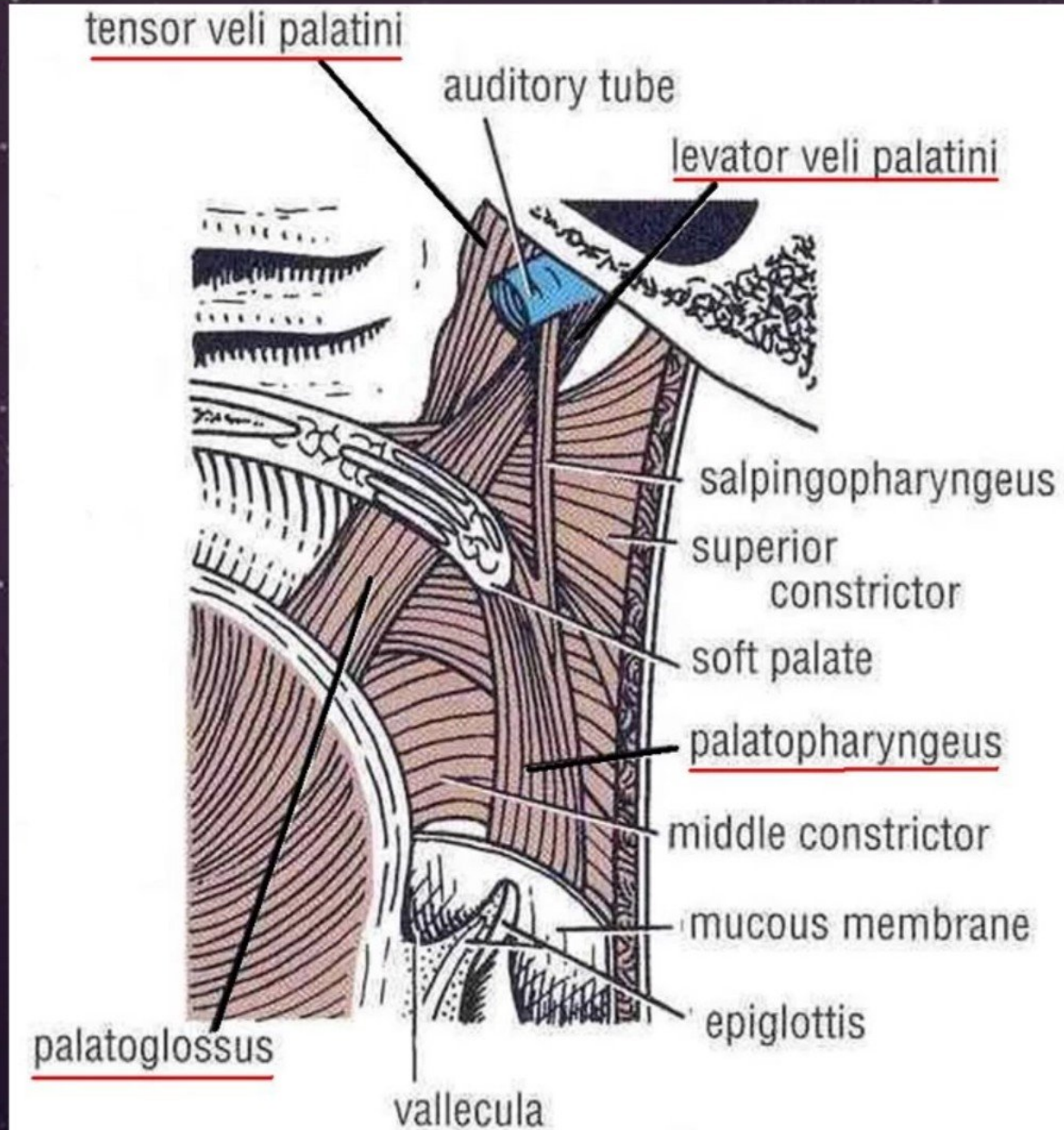
✧ **Action:** pulls root of tongue upward, narrowing oropharyngeal isthmus

& Palatopharyngeus

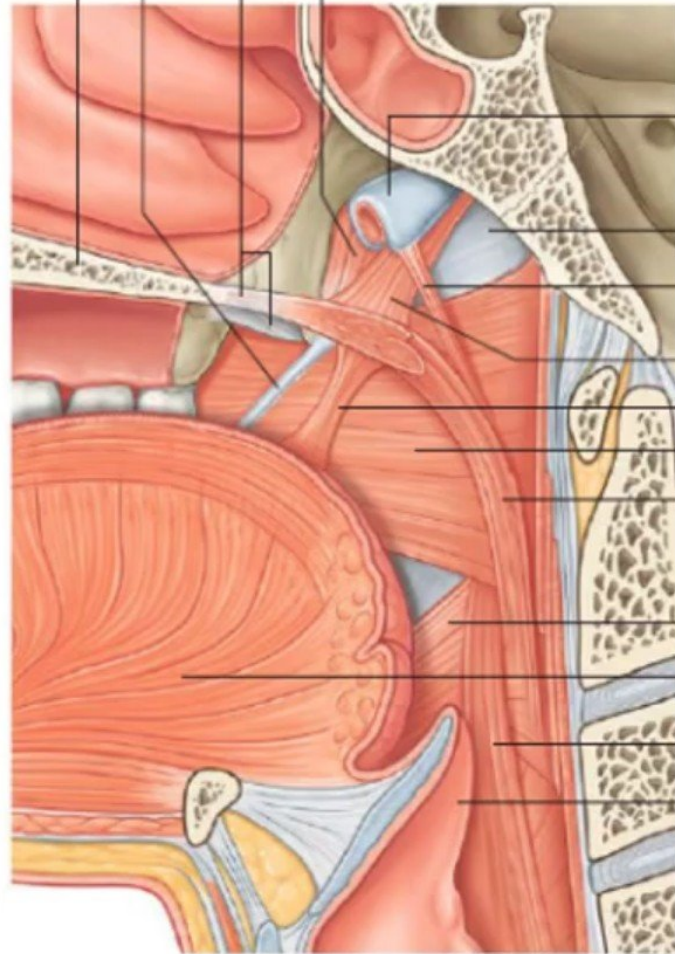
✧ **Origin:** palatine aponeurosis

✧ **Insertion:** posterior border of thyroid cartilage

✧ **Action:** Elevates wall of the pharynx



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Pharyngobasilar fascia

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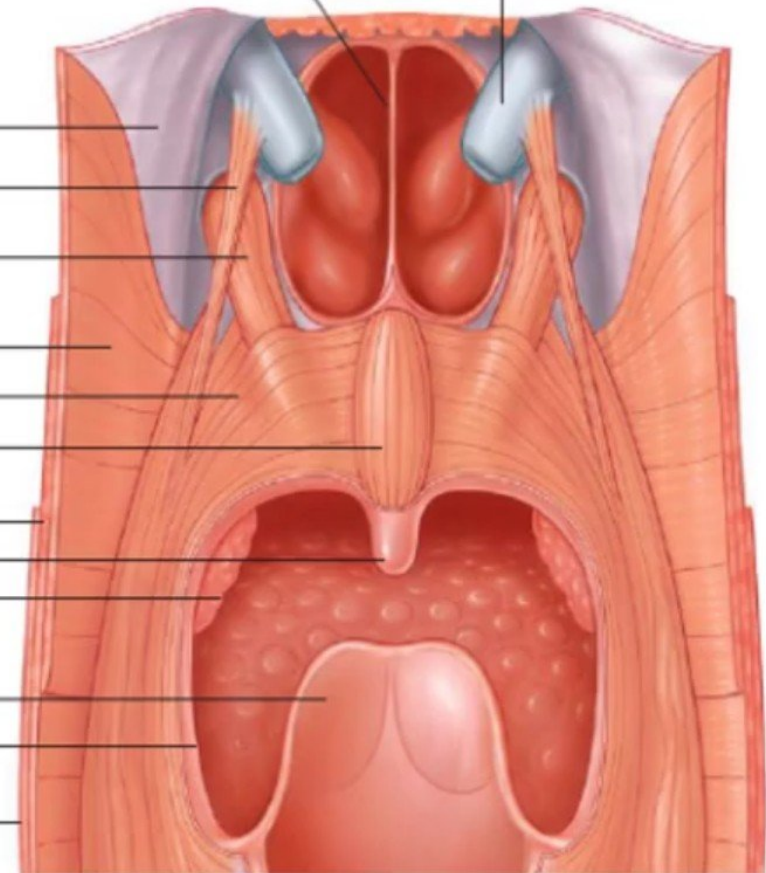
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Muscles of the soft palate (sagittal section)

Muscles of the soft palate (posterior view)

Sensory Nerve Supply

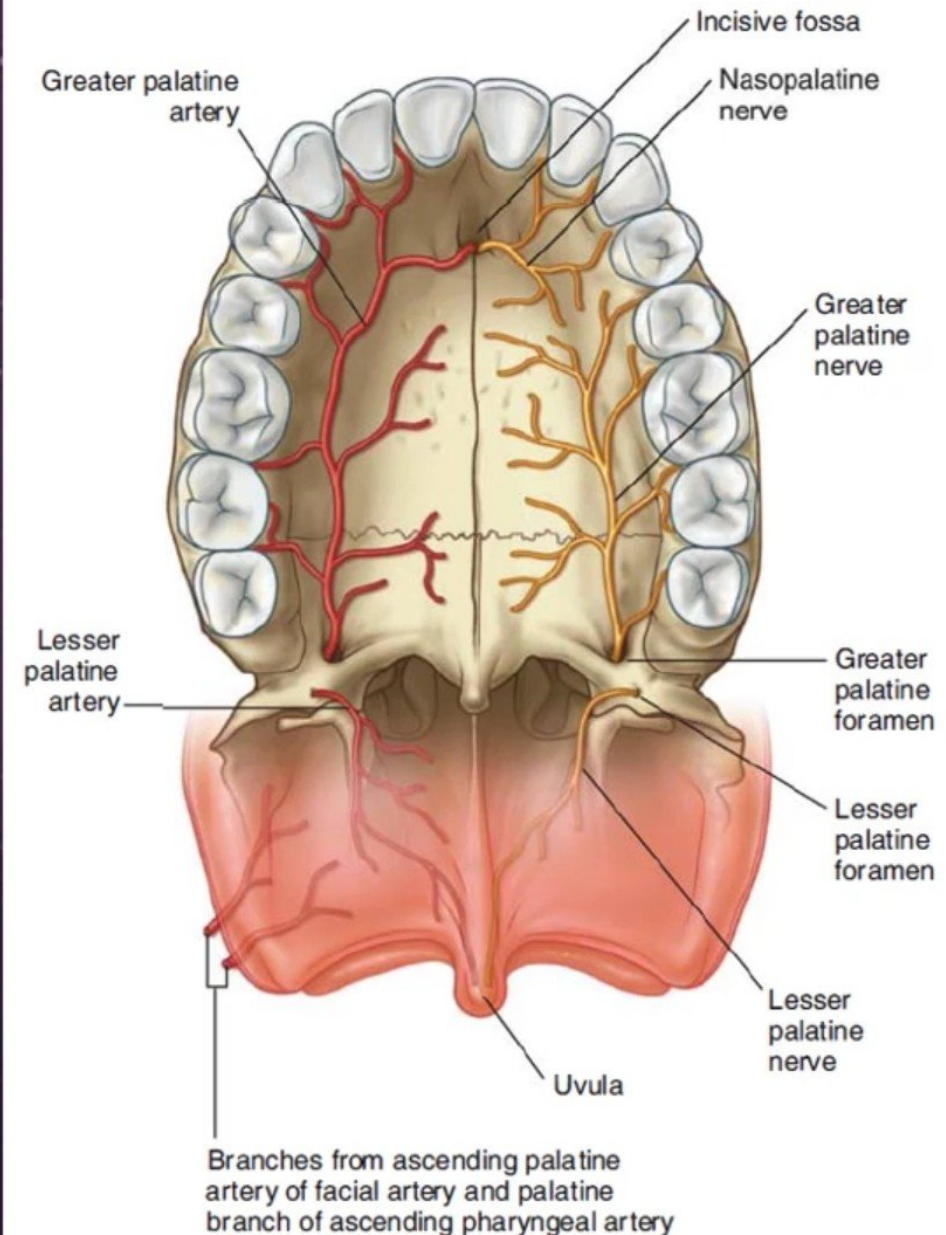
↳ Mostly by the **maxillary** nerve through its branches:

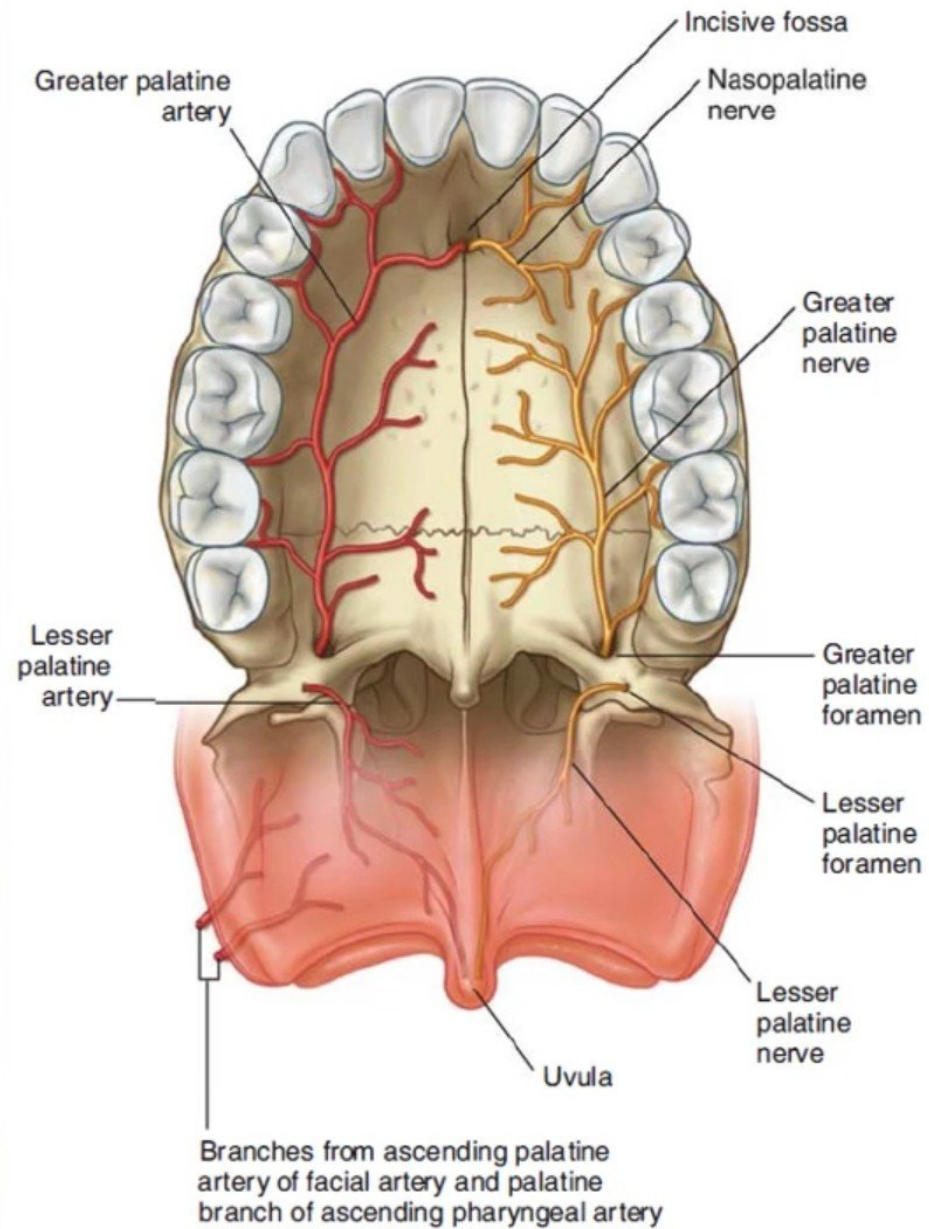
✧ Greater palatine nerve

✧ Lesser palatine nerve

✧ Nasopalatine nerve

↳ **Glossopharyngeal** nerve supplies the region of the soft palate





Motor Nerve Supply

↳ All the muscles, are supplied by the:

⌘ Pharyngeal plexus

except **tensor veli palatini**,

↳ **Tensor veli palatini** supplied by the:

⌘ Nerve to medial pterygoid, a branch of the mandibular division of the **trigeminal nerve**

Blood Supply

↳ Branches of the maxillary artery

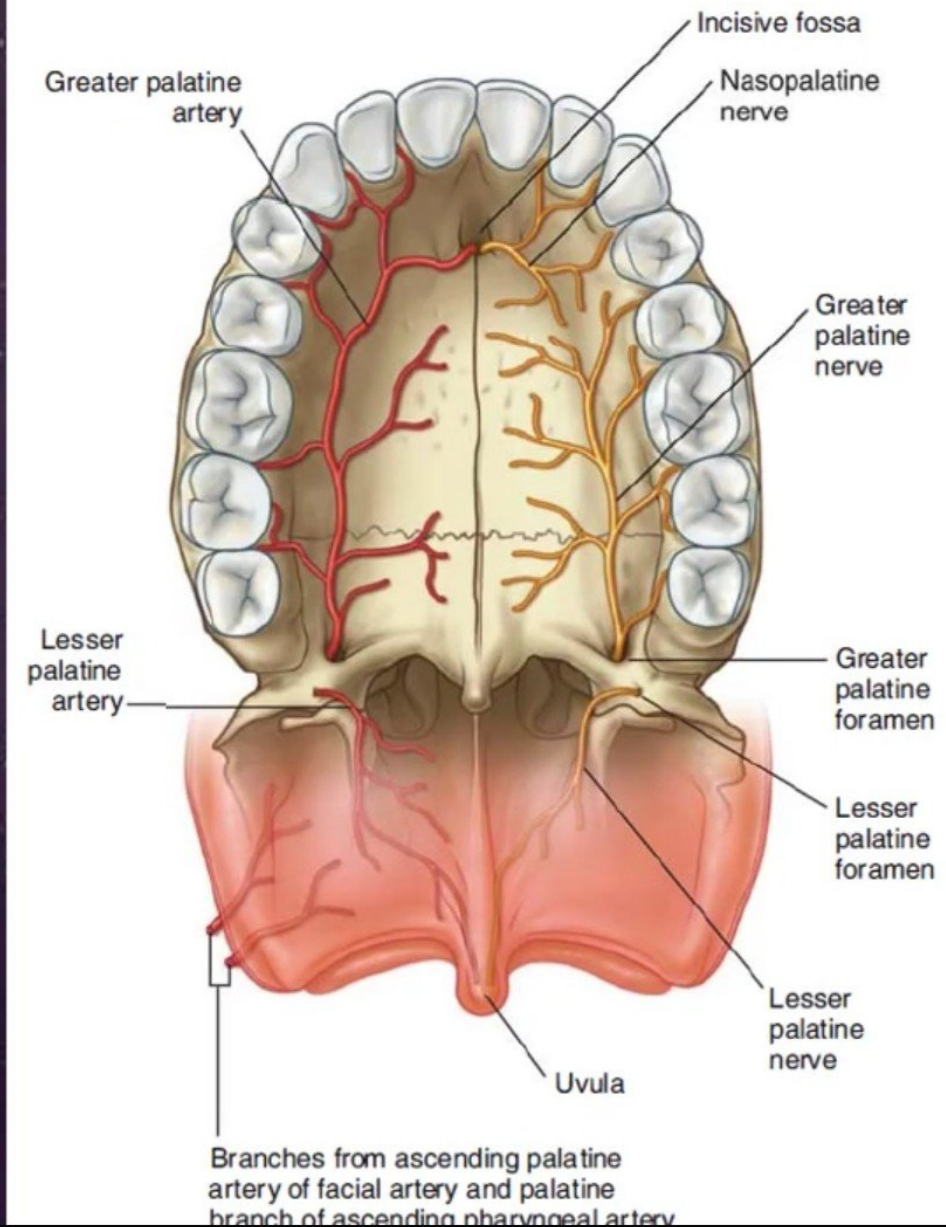
✧ Greater palatine

✧ Lesser palatine

✧ Sphenopalatine

↳ Ascending palatine, branch of the facial artery

↳ Ascending pharyngeal, branch of the external carotid artery



Clinical Notes

❧ Cleft palate:

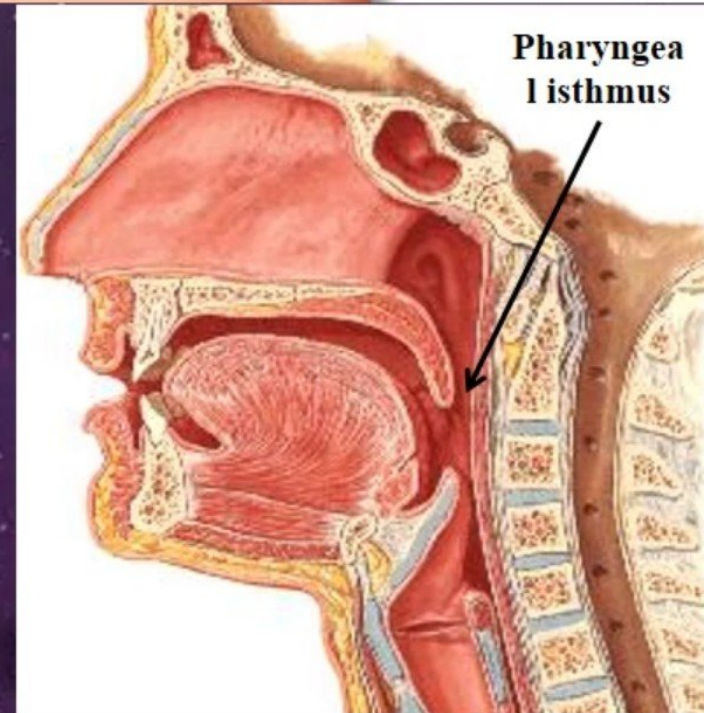
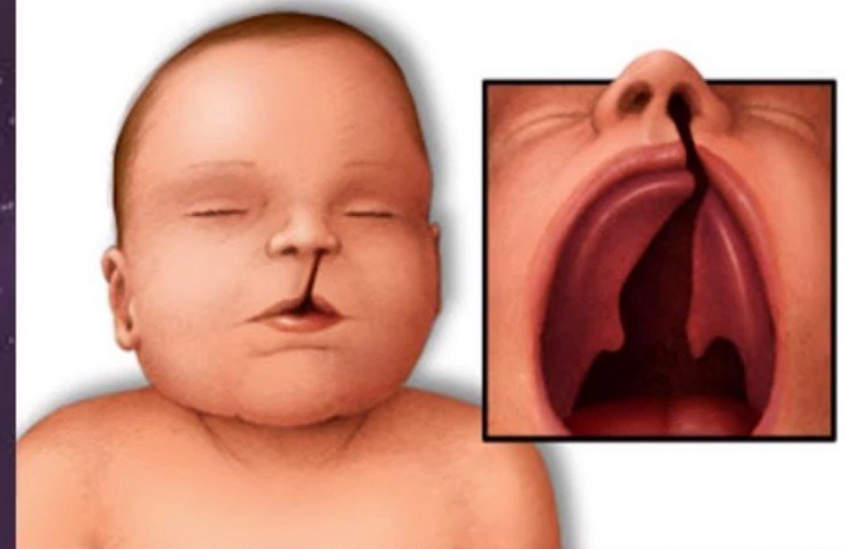
❧ Unilateral

❧ Bilateral

❧ Median

❧ Paralysis of the soft palate

❧ The **pharyngeal isthmus** can not be closed during swallowing and speech



Thanks